

Motivation

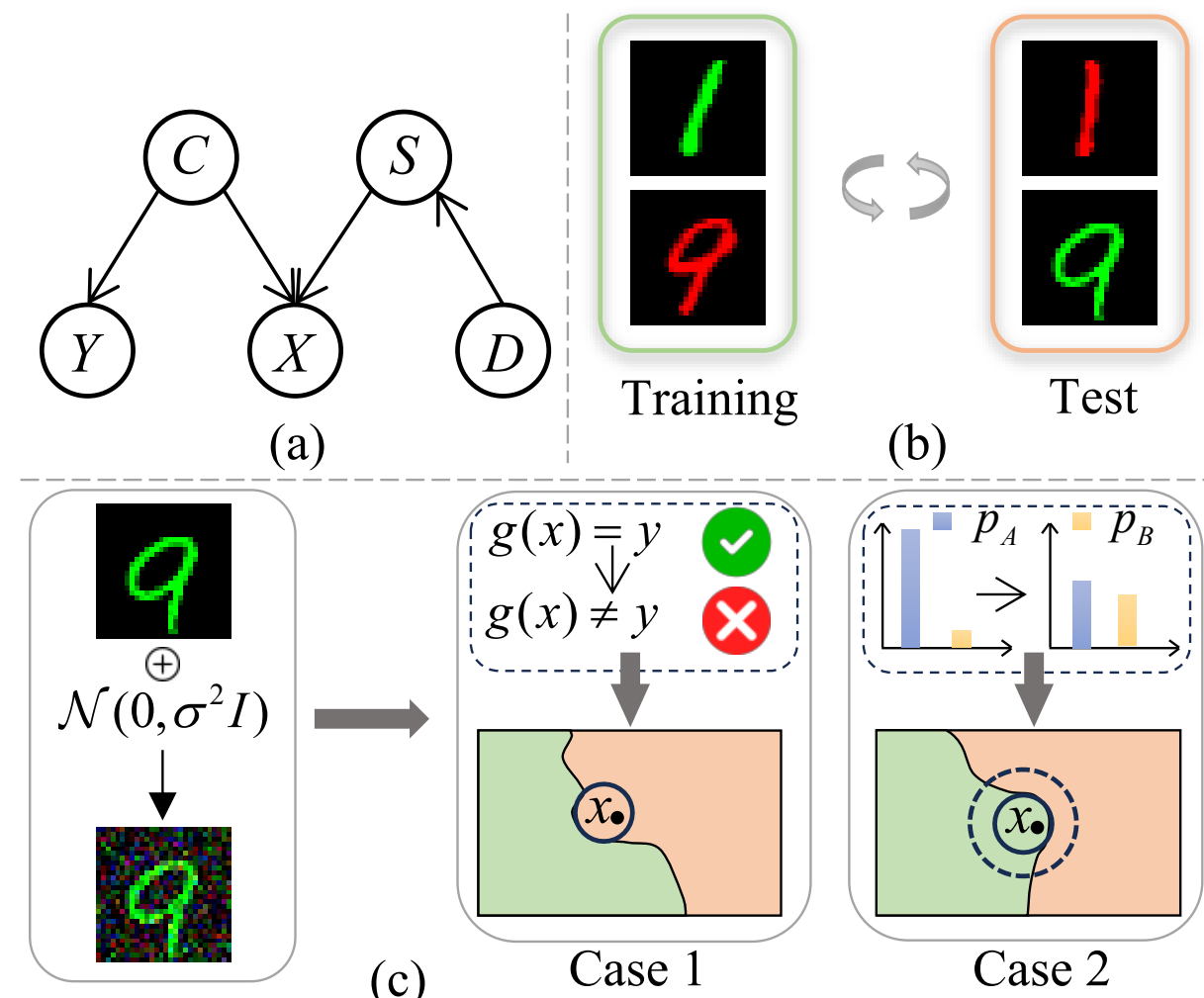
- **Certified defense** is known for its guarantees against arbitrary adversarial perturbations within a range. However, existing works based on this lack **generalizability** in other data domains with distribution shifts.
- Finding **causal factors** can eliminate spurious correlations and further enhance generalizability.
- In this work, we aim to **design a model that can generalize its robustness across domains incorporating a causal perspective**.

Table: Comparison of the certified defense performance with/without domain shift. The metrics include the prediction accuracy and the Average Certified Radius (ACR).

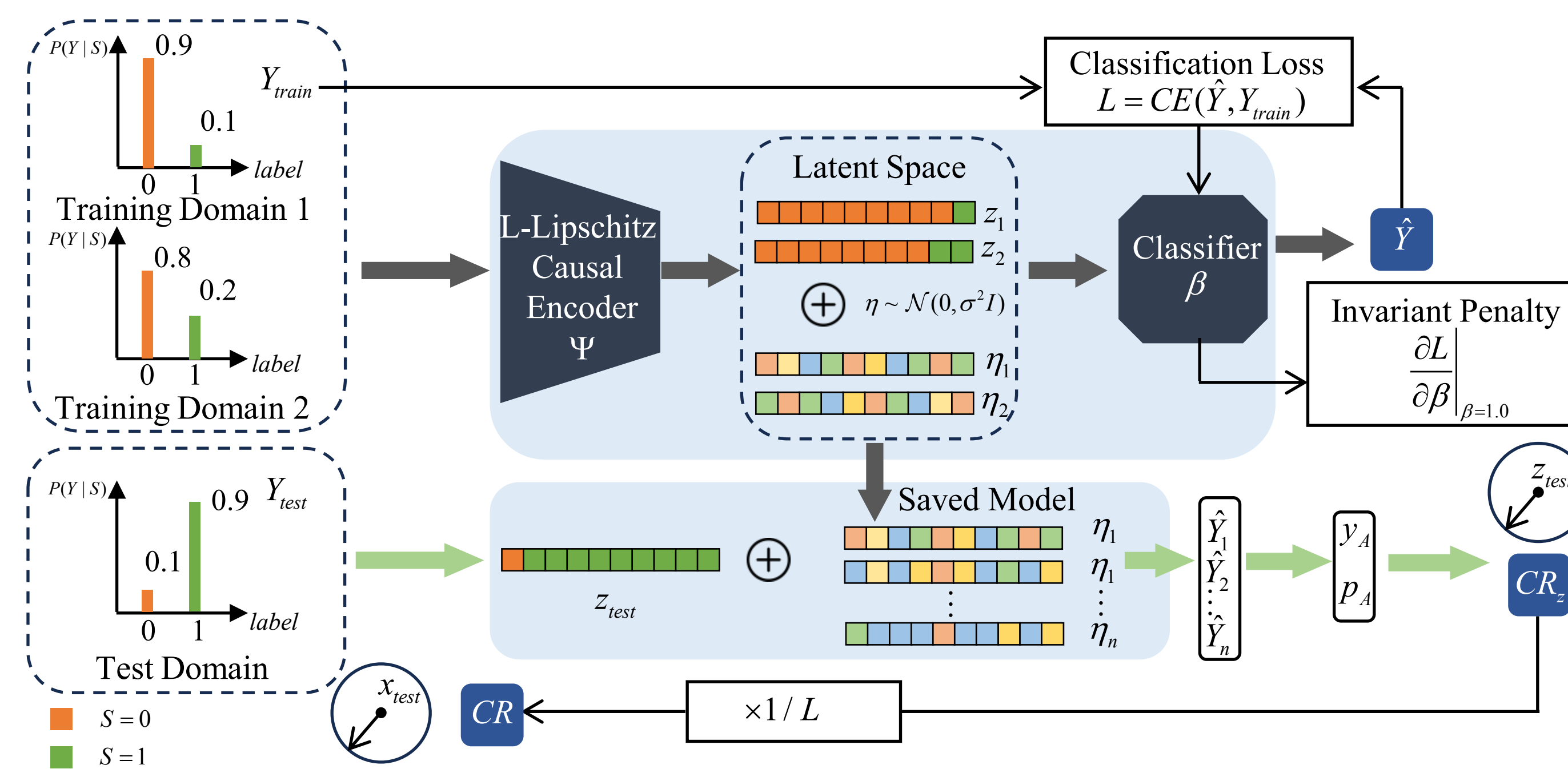
Dataset	Test Acc	ACR ($\sigma = 0.25$)
CMNIST	21.01%	0.07
MNIST	72.03%	0.37

Background & Challenges

- **Spurious Correlation:** Spurious correlations varying across domains adversely affect robustness.
- **Distribution Shift:** Distribution shifts frequently pose difficulties for the model in defending against perturbations on factors that causally determine the label in unseen domains.
- **Theoretical Guarantee:** It is important to provide theoretical guarantees for robustness in other data domains. However, most existing works remain in empirical observations.



Methodology



Causal Invariance Learning Module

- Causal Invariance Assumption

$$P^d(Y|C) = P^{d'}(Y|C), \forall d, d' \in \mathcal{D}$$

- Robustness Perspective

$$P^{d'}(Y|X) = \int_{c \in \mathcal{C}} P^{d'}(c|X) P^{d'}(Y|c) = \int_{c \in \mathcal{C}} P^{d'}(c|X) P^d(Y|c)$$

- Invariant Risk Minimization

$$\mathcal{L}_{\text{IRM}} = \sum_{d \in \mathcal{D}_{\text{tr}}} R^d(\beta \circ \Psi) + \lambda \cdot \|\nabla_{w|w=1.0} R^d(w(\beta \circ \Psi))\|^2$$

Theoretical Guarantee

- Certified Radius in Latent Space

$$CR_z(\beta; x, y) = \frac{\sigma}{2} (\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B))$$

- Certified Radius in Image Space

$$CR \geq CR_z / L$$

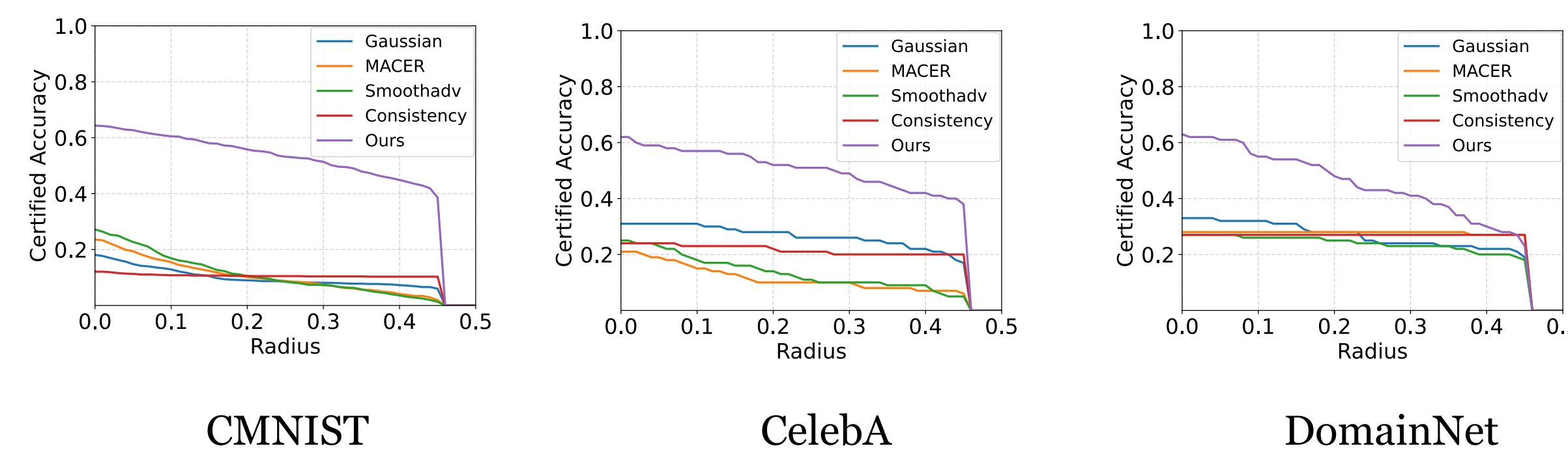
Experiments

Overall performance

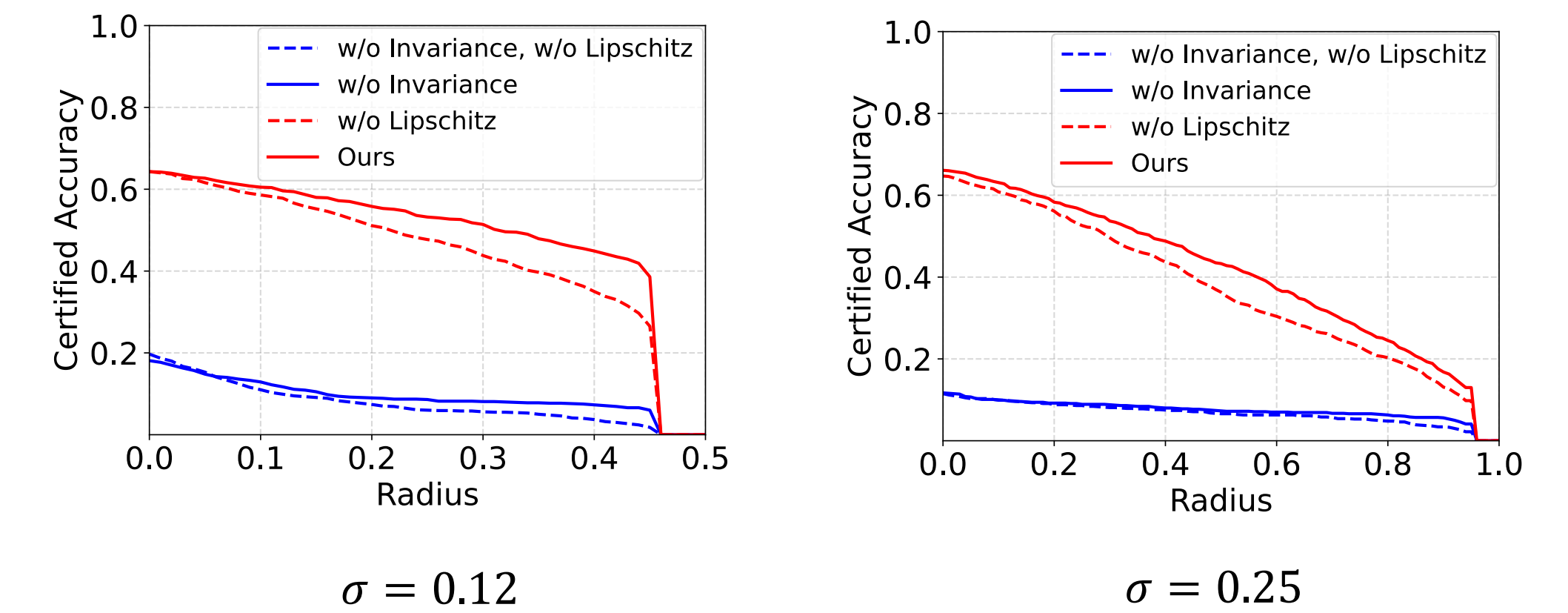
Table: The performance of different certified defense methods

Datasets	Models	$r = 0.00$	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	ACR
CMNIST	Gaussian	18.1	14.8	12.9	10.5	9.0	8.6	8.1	7.8	7.3	6.0	0.0458
	MACER	23.6	19.4	15.5	12.4	10.1	8.8	7.4	5.6	4.1	1.9	0.0482
	SmoothAdv	27.2	22.7	16.9	13.6	10.5	8.5	7.3	5.8	3.6	1.3	0.0518
	Consistency	12.1	11.3	10.8	10.7	10.6	10.5	10.4	10.4	10.3	10.3	0.0488
	GLEAN (Ours)	64.3	62.7	60.5	58.0	55.8	53.2	51.4	47.9	44.9	38.6	0.2466
CelebA	Gaussian	31.0	31.0	31.0	29.0	28.0	26.0	26.0	24.0	22.0	17.0	0.1218
	MACER	21.0	19.0	15.0	13.0	10.0	10.0	10.0	8.0	7.0	6.0	0.053
	SmoothAdv	25.0	23.0	18.0	16.0	14.0	11.0	10.0	9.0	9.0	5.0	0.0623
	Consistency	24.0	24.0	23.0	23.0	22.0	21.0	20.0	20.0	20.0	20.0	0.0989
	GLEAN (Ours)	62.0	59.0	57.0	56.0	52.0	51.0	49.0	45.0	42.0	38.0	0.2326
DomainNet	Gaussian	33.0	32.0	32.0	31.0	28.0	25.0	24.0	23.0	22.0	19.0	0.1223
	MACER	28.0	28.0	28.0	28.0	28.0	28.0	28.0	28.0	27.0	27.0	0.1272
	SmoothAdv	27.0	27.0	26.0	26.0	25.0	24.0	23.0	23.0	20.0	18.0	0.1097
	Consistency	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	0.1234
	GLEAN (Ours)	63.0	61.0	55.0	54.0	48.0	43.0	41.0	37.0	30.0	23.0	0.2088

Figure: Comparison of certified accuracy obtained by three different methods



Ablation study



Parameter study

